

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The financial landscape of Zimbabwe has undergone a radical transformation over the last two decades, driven primarily by recurring liquidity challenges and the rapid proliferation of telecommunications infrastructure. In an environment where physical currency is often scarce, mobile money platforms have emerged as the primary vehicle for financial inclusion. Official figures indicate that by 2020, active mobile money subscriptions in Zimbabwe reached approximately 7.6 million, with Econet Wireless's EcoCash platform dominating the market and facilitating urban-to-rural remittances.

At the University of Zimbabwe, this digital shift is encapsulated in the "Emhare" administrative and student registration system. This centralized portal integrates bank payments, mobile money wallets, and institutional ERP systems to facilitate the high-volume collection of tuition fees. However, this systemic reliance on digital payments has introduced a heightened risk of cyber-financial crime. The convergence of telecommunications and banking has created security loopholes where sophisticated fraudsters utilize techniques such as social engineering, SIM swap fraud, and smurfing—splitting large illicit transfers into smaller, less suspicious amounts—to evade detection.

Traditional rule-based fraud detection systems, which rely on rigid thresholds, are increasingly incapable of countering these dynamic threats. Consequently, there is an urgent mandate under Zimbabwe's Education 5.0 policy—which emphasizes innovation and industrialization—for universities to develop intelligent, machine-learning-based frameworks that can autonomously identify and mitigate fraudulent activities within their financial ecosystems.

1.2 Problem Statement

The University of Zimbabwe relies heavily on the Emhare system for mobile money transactions, yet there is no dedicated automated system to detect fraudulent activities associated with these transactions in real-time. Current verification processes are largely manual, making them imprecise, costly, and too slow to prevent financial loss. Fraud incidents, including phishing and unauthorized account access, lead to significant institutional financial losses and reduced trust in the university's digital payment systems. Without an intelligent detection framework, the institution remains vulnerable to evolving cyber-financial crimes that bypass traditional manual audits.

1.3 Aim of the Study

To design and develop an intelligent Mobile Money Fraud Detection System using eXtreme Gradient Boosting (XGBoost) to detect and flag fraudulent transactions within the University of Zimbabwe ecosystem.

1.4 Objectives of the Study

The study is guided by the following objectives:

1. To analyze common mobile money fraud patterns specific to the Zimbabwean socio-economic context.
2. To review existing machine learning techniques and their efficacy in detecting financial anomalies.
3. To design and implement a fraud detection model using the XGBoost algorithm.
4. To evaluate the system's performance using balanced metrics such as Precision, Recall, and F1-score.

1.5 Scope of the Study

The study focuses on detecting fraud in mobile money transactions within the University of Zimbabwe's financial ecosystem. It utilizes the synthetic PaySim dataset, which models realistic transaction behaviors and fraud scenarios from a real African mobile money operator, as a proxy for institutional data due to privacy constraints.

1.6 Significance of the Study

This research aligns with the national Education 5.0 mandate by shifting from theoretical research to the creation of industrial solutions. The proposed system will:

1. **Enhance Institutional Security:** Provide real-time protection for the Emhare payment portal.
 2. **Reduce Financial Loss:** Minimize revenue leakages from tuition fraud and chargebacks.
 3. **Promote Digital Trust:** Safeguard student financial interactions, ensuring a secure digital campus environment.
-

CHAPTER 2: LITERATURE REVIEW AND FEASIBILITY STUDY

2.1 Review of Related Literature

2.1.1 Dal Pozzolo et al. (2015)

Dal Pozzolo et al. investigated the challenge of concept drift, where fraud patterns change over time due to seasonal shifts or evolving attacker tactics. The study proposed training separate classifiers on immediate investigator feedback and delayed labels to improve alert precision. This is critical for systems where transaction confirmation may be delayed by several days.

2.1.2 Lopez-Rojas, Elmir, and Axelsson (2016)

The authors introduced PaySim, a multi-agent simulation framework that generates realistic synthetic data. By using aggregated data from an African mobile money operator, PaySim creates a rich environment for testing algorithms without violating data privacy laws, modeling the behavior of both "good" agents and fraudsters.

2.1.3 Whitrow et al. (2009)

This study demonstrated that transaction aggregation—summarizing account behavior over time windows (e.g., previous 24 hours)—provides better predictive power than analyzing isolated transactions. Aggregation reveals subtle deviations from a user's historical baseline, which is essential for detecting account compromise.

2.1.4 Bahnsen et al. (2013)

Bahnsen et al. focused on cost-sensitive decision trees, arguing that misclassifying a fraud event (false negative) is significantly more expensive than a false alarm (false positive). Their approach incorporates instance-dependent costs into the model training to minimize overall financial impact.

2.1.5 Carcillo et al. (2021)

This comprehensive review introduced the Operational Capability Triad (ORT), focusing on data realism, resilience to drift, and human-in-the-loop interpretability. The study emphasized that gradient-boosted decision trees remain the strongest models for tabular financial data.

2.1.6 Chen and Guestrin (2016)

Chen and Guestrin introduced XGBoost, an eXtreme Gradient Boosting system known for its scalability and speed. Its regularized objective function prevents overfitting, making it uniquely suited for the sparse and imbalanced nature of financial logs.

2.1.7 Phua et al. (2010)

Phua et al. provided a theoretical hierarchy for selecting data mining methods, emphasizing classification for known patterns and clustering for identifying new, anomalous behaviors. They highlighted the shift from single-record checks to relational learning.

2.1.8 Makori et al. (2019)

Makori identified mobile money vulnerabilities specific to Sub-Saharan Africa, such as social engineering and fake transaction notifications. This underscores the need for a localized detection system at the University of Zimbabwe that accounts for regional USSD and P2P transaction patterns.

2.1.9 Dal Pozzolo et al. (2017)

In a follow-up study, the authors emphasized the superiority of real-time streaming frameworks over batch-processing models. They proposed scalable frameworks like SCARFF for handling massive streams of financial data in near-real time.

2.1.10 University of Zimbabwe Institutional Repository (2020)

Institutional research by Tachiona and others proposed conceptual frameworks for detecting financial crimes, aligning research with national security and the Education 5.0 paradigm. Their work supports the development of localized solutions tailored to the university's specific transaction architectures.

2.2 Feasibility Study

2.2.1 Technical Feasibility

The University of Zimbabwe's Department of Computer Engineering provides the necessary intellectual foundations, including expertise in AI and database engineering. The system requires Python 3.x with industry-standard libraries such as XGBoost and Scikit-learn for handling imbalanced datasets. GPU-accelerated servers are required to handle millions of PaySim records and achieve sub-second inference latency.

2.2.2 Economic Feasibility

The initial investment in hardware and personnel is offset by the potential for substantial loss prevention. Reducing tuition fraud and administrative investigation costs ensures a high long-term return on investment (ROI) for the university, as catch rates for fraud can be improved significantly compared to manual methods.

2.2.3 Legal and Regulatory Feasibility

The system must comply with the "Cyber and Data Protection Act [Chapter 12:07]," enacted in 2021. Key requirements overseen by POTRAZ include:

- **Data Minimization:** Only attributes necessary for classification should be stored.
- **Breach Notification:** Mandatory 24-hour window for reporting data breaches to POTRAZ.
- **Licensing:** Organizations processing data of more than 50 individuals must obtain a data controller's license from POTRAZ by March 2025.

2.2.4 Operational Feasibility

Operational viability depends on integrating the XGBoost engine as middleware within the Emhare portal. The system will provide investigators with "Explainable AI" metrics (e.g., SHAP values) to visualize the rationale behind alerts, ensuring the solution is integrated into the daily workflows of the bursary department.

2.3 References

1. G. Dal Pozzolo et al., "Adversarial drift detection," IEEE, 2015.
2. E. Lopez-Rojas et al., "PaySim: A financial mobile money simulator," 2016.
3. C. Whitrow et al., "Transaction aggregation for fraud detection," IEEE, 2009.
4. A. Bahnsen et al., "Cost-sensitive decision trees," IEEE, 2013.
5. F. Carcillo et al., "Fraud detection review," IEEE, 2021.
6. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," ACM, 2016.
7. C. Phua et al., "A comprehensive survey of data mining-based fraud detection," 2010.
8. J. Makori et al., "Mobile money fraud in Africa," 2019.
9. G. Dal Pozzolo et al., "Real-time fraud detection," IEEE, 2017.
10. University of Zimbabwe, "Mobile financial crime detection framework," 2020.